



Supporting Online Material for

The Connectivity Map: Using Gene-Expression Signatures to Connect Small Molecules, Genes, and Disease

Justin Lamb,^{*} Emily D. Crawford, David Peck, Joshua W. Modell, Irene C. Blat, Matthew J. Wrobel, Jim Lerner, Jean-Philippe Brunet, Aravind Subramanian, Kenneth N. Ross, Michael Reich, Haley Hieronymus, Guo Wei, Scott A. Armstrong, Stephen J. Haggarty, Paul A. Clemons, Ru Wei, Steven A. Carr, Eric S. Lander, Todd R. Golub^{*}

^{*}To whom correspondence should be addressed. E-mail: golub@broad.harvard.edu,
justin@broad.mit.edu

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Materials and Methods
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References

Other Supporting Online Material for this manuscript includes the following:
(available at www.sciencemag.org/cgi/content/full/313/5795/1929/DC1)

Table S1, Signatures S1 to S11, Results S1 to S11 as zipped archive
[1132939s.zip](#)

Materials and Methods

Treatments

Cells (MCF7, PC3, HL60, SKMEL5) were treated in batches composed either of four small-molecule treatments and one vehicle control (6-well dishes; development) or forty-two treatments and six vehicle controls (96-well plates; production). Treatments were made 24 h after plating. Full details of cell lines, small molecules and treatment conditions for the dataset (build 01) are provided as table S1.

Gene Expression Profiling

Total RNA was isolated using TRIzol Reagent (Invitrogen) either as directed by the manufacturer (development batches) or in a miniaturized and semi-automated microtiter plate format based on the manufacturer's instructions (production batches). Synthesis of cRNA target, its hybridization to microarrays and scanning of those arrays was performed using Affymetrix GeneChip products and reagents, in accordance with the manufacturer's recommendations. Total RNA from development batches was processed manually using HG-U133A cartridge arrays (part number 510681). Total RNA from production batches was amplified and labeled using the GeneChip Array Station (GCAS) and hybridized to HT_HG-U133A (early access version; part number 520276) High-Throughput Arrays (HTA) which were scanned using the HT Scanner, allowing 96 samples to be processed in parallel with minimal operator intervention. Scans failing to satisfy basic data quality criteria were discarded.

The data discussed in this publication have been deposited in NCBI's Gene Expression Omnibus (GEO, <http://www.ncbi.nlm.nih.gov/geo/>) and are accessible through GEO series accession number GSE5258. Data are also available for download at <http://www.broad.mit.edu/cmap>.

Data Preprocessing

Each treatment instance (*ie* one treated *versus* vehicle control pair) was represented by a non-parametric rank-ordered list of all probe sets on the HG-U133A array (22,283), constructed as follows. First, scan data were converted to average difference values and confidence calls with MAS 5.0 (Affymetrix). Next, data content from production batches was made equivalent to that from development batches by discarding expression values from the 676 registration probe sets unique to HTA and setting the average difference values and confidence calls for the fifteen probe sets missing from these arrays to 0 and 'A', respectively. Average difference values from treatment scans were then scaled relative to those from their corresponding vehicle control using a linear-fit-on-P-call algorithm. In production batches, control average difference values were set to the arithmetic mean of the values from the six individual control scans, and a probe set was given a 'P' call if more than three of the six confidence calls were

'P'. Next, average difference (control) and scaled average difference (treatment) values less than a primary threshold value (20, development; 50, production) were set to that threshold value. Finally, probe sets were ranked in descending order of d , where d is the ratio of the corresponding treatment-to-control values. For probe sets where $d=1$, a lower threshold (2, development; 5, production) was applied to the original (scaled) average difference values and a new treatment to control ratio (d') calculated. These probe sets were then sub-sorted in descending order of d' .

Signatures

Gene-expression signatures were represented by two sets of HG-U133A probe sets ('up tags' and 'down tags'). Gene identifiers from external signatures were mapped to HG-U133A probe sets using UniGene, HomoloGene (both NCBI) and NetAffx or GeneChip Annotation Files (both Affymetrix), as required. All mapped probe sets were included where there was a one-to-many relationship. Internal signatures were populated with probe sets having amplitudes above and below given values in all selected query instances. Amplitude a is defined as $(u-v)/((u+v)/2)$, where u is the thresholded scaled average difference value (treatment) and v is the thresholded average difference value (vehicle). Tag lists for all signatures are provided as Signatures S1-S11.

Connectivity Analytics

Treatment instances were rank ordered with respect to a given query signature using a gene-set enrichment metric based on the Kolmogorov-Smirnov statistic (1, 2), as follows. For each instance i , compute an enrichment score for the set of probe sets ('tags') representing the up- or down- regulated genes in the signature; ks_{up}^i and ks_{down}^i , respectively. Let n be the total number of probe sets (22,283) and t be the number of tags. Construct a vector V of the position (1... n) of each tag in the ordered list of all probe sets (see Data Preprocessing, above) and sort these components in ascending order such that $V(j)$ is the position of tag j , where $j = 1, 2, \dots, t$. Compute the following two values:

$$a = \max_{j=1}^t \left[\frac{j}{t} - \frac{V(j)}{n} \right]$$

$$b = \max_{j=1}^t \left[\frac{V(j)}{n} - \frac{(j-1)}{t} \right]$$

... and set $ks^i = a$, if $a > b$ or $ks^i = -b$ if $b > a$. The connectivity score S^i is set to zero where ks_{up}^i and ks_{down}^i have the same algebraic sign. Otherwise, set $s^i = ks_{up}^i - ks_{down}^i$, $p = \max(s^i)$ and $q = \min(s^i)$ across all instances. The connectivity score S^i for the non-zero instances is defined as s^i / p where $s^i > 0$, or $-(s^i / q)$ where $s^i < 0$. Instances are then ranked in descending order of S and ks_{up} . Finding multiple independent instances of the same perturbagen with high (or

low) rankings was taken to indicate positive (or negative) connectivity between that perturbation and the biology represented in the query signature.

The significance of a particular distribution of a set of instances in the ordered list of all instances was estimated by permutation, as follows. The Kolmogorov-Smirnov statistic is computed for the set of t instances of interest in the ordered list of n instances as above, giving an enrichment score KS_0 . Then, for each of r trials, select t instances at random from the set of n instances and compute KS_m , and count the number of times s that $|KS_m| \geq |KS_0|$ is true. The frequency of this event (s / r) can be taken as a (two-sided) p-value. We set $r=10,000$.

Supplementary Figures

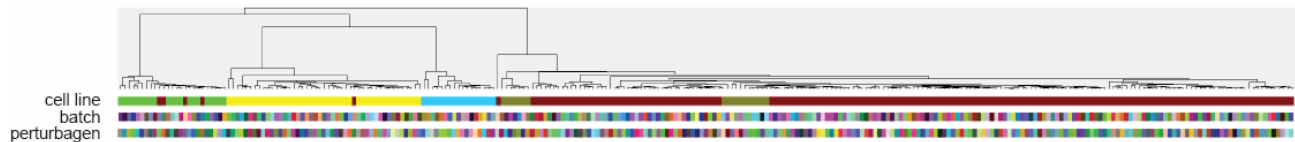


Fig. S1. Hierarchical Clustering Finds Few Small Molecule Connections. Dendrogram showing the similarity between treatments. Colors represent cell lines, batches and perturbagens. The 271 treatment scans from development batches were scaled relative to one another using RMA with quantile normalization (3). Probe sets reporting ≥ 20 and $\leq 16,000$ units, and with ≥ 5 fold variation across the dataset were selected. Samples were hierarchically clustered in the space of these 4,800 probe sets using a Pearson distance metric and average linkage.

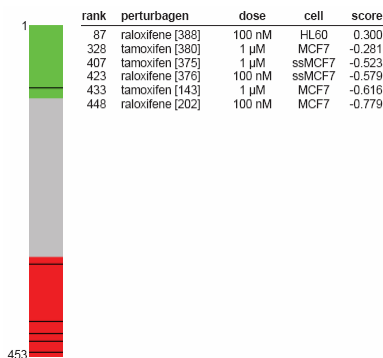


Fig. S2. Tamoxifen and Raloxifene. Barview showing all tamoxifen ($n=3$) and raloxifene ($n=3$) instances for the 17β -estradiol signature. See Fig. 3B for comparison.

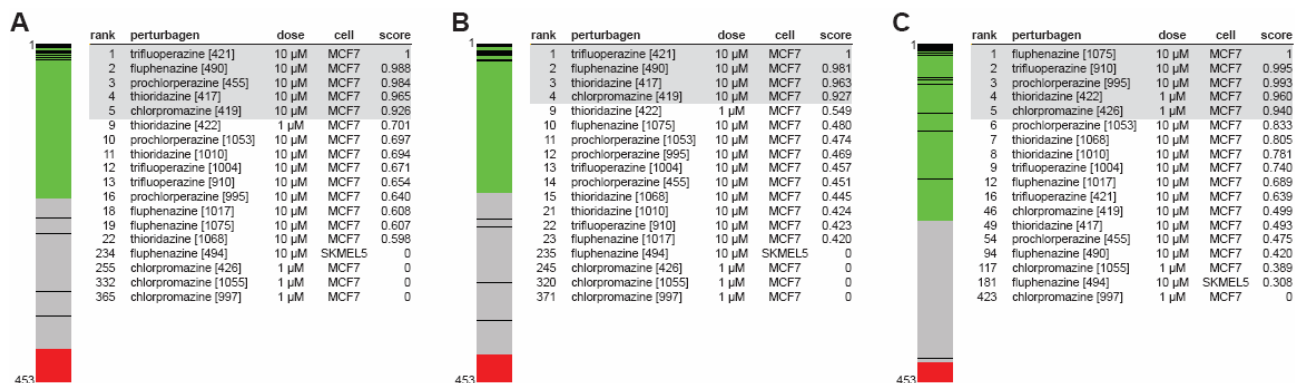


Fig. S3. Alternate Phenothiazine Signatures. (A-C) Barviews showing all thioridazine ($n=4$), chlorpromazine ($n=4$), fluphenazine ($n=4$), trifluoperazine ($n=3$) and prochlorperazine ($n=3$) instances for alternate internal phenothiazine signatures (Signatures S4-S6, respectively). The instances used to generate the

signatures are shaded. See Fig. 4B for comparison. Unabridged results from these queries are provided as Results S4-S6.

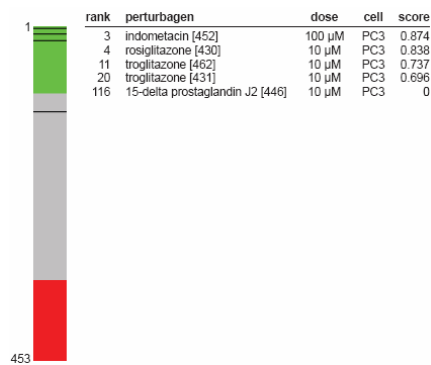


Fig. S4. PPAR γ Agonists are Connected with Diet-induced Obesity in Rats. Barview (as Fig. 2) showing all instances of troglitazone ($n=2$), rosiglitazone ($n=1$), indometacin ($n=1$) and 15-delta prostaglandin J2 ($n=1$) in PC3 cells. Unabridged results from this query are provided as Result S8.

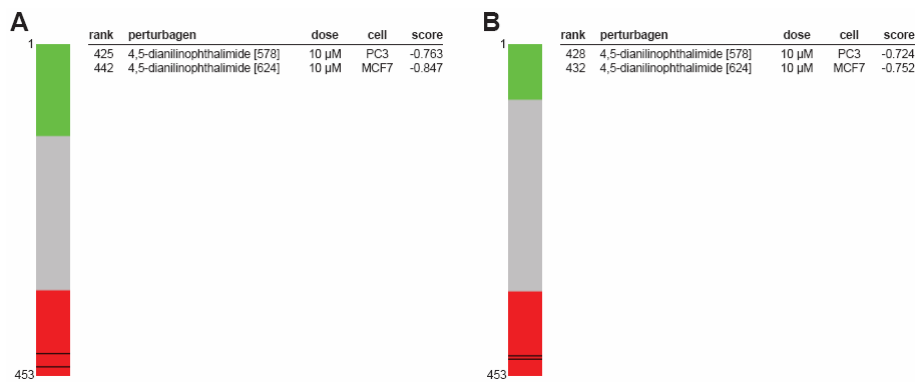


Fig. S5. 4,5-dianilinophthalimide is Negatively Connected with Alzheimer's Disease. (A) Barviews (as Fig. 2) showing the two instances of 4,5-dianilinophthalimide with the Hata *et al* (4) signature, and (B) the Ricciarelli *et al* (5) signature. Permutation p -values for the set of 4,5-dianilinophthalimide instances are 0.0075 and 0.0057, respectively. Unabridged results from these queries are provided as Results S9 and S10.

Supplementary References

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